

# Matthew Snelgrove

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matthewSnelgrove  
portfolio

## Education

**University of Toronto, Scarborough | Computer Science | CGPA: 3.72** September 2021 – Present

- Courses: Data Structures, Algorithms, Software Design, Security, Databases, Web Development

## Experience

**SS&C Technologies (IFDS) | Software Developer** September 2023 – April 2024

- Eliminated 3 hours of managerial work per week by automating component dependency tracking with internal APIs, and visualized output in graphical and tabular views in a modern React web app.
- Ensured the maintainability of the project and expedited its handover by documenting the codebase with jsDoc and creating comprehensive wiki entries.
- Improved system reliability by identifying and patching production bugs across team apps rapidly.
- Accelerated modernization projects by investigating internal tools and delivering actionable user guides to streamline their integration into existing systems.

**Ontario Ministry of Transportation | Junior Technical Analyst** January 2023 – April 2023

- Developed RESTful API for modernization of critical driver/vehicle registration systems with Java Spring Boot.
- Reduced debugging time by over 50% by performing automated unit and integration testing with JUnit and Postman to verify functionality and identify errors caused by frequently changing internal dependencies.
- Recovered ~3 hours per week of developer time by building and deploying new versions with Maven and Azure Pipelines instead of time-consuming and error-prone manual builds.

## Projects

**Superbidido**   October 2024 – December 2024

- Implemented core features for the auction-based marketplace app by developing a flexible and scalable RESTful API using Express and a relational database using PostgreSQL.
- Minimized cloud costs and optimized server performance by handling image storage and retrieval with Google Cloud Storage and integrating lifecycle rules to delete orphaned files automatically.
- Deployed app with a three-tier architecture on a Google Cloud VM using Docker, enabling targeted optimization, independent updates to each layer, and robust scalability.
- Ensured reliability, security, and ample performance at minimal cost by customizing a Google Cloud VM to host the containerized application, setting up monitoring and alerts, and enforcing secure network configurations.
- Automated consistent and reliable deployments with a bash script to build images, upload them to the VM, and start containers with secrets using a Docker Compose file.
- Fully secured the app and eliminated web vulnerabilities with helmet.js, HTTPS, data validation, CSRF protection, and session-based authentication.

**Stock Portfolio App**  August 2024

- Designed and implemented a PostgreSQL database for the social stock tracking platform.
- Optimized statistical correlation and covariance queries by precomputing the results in a materialized view, reducing query time by 98%.

## Technical Skills

**Languages:** Java, JavaScript/TypeScript, Python, C/C++, C#, SQL (Postgres), HTML, CSS

**Frameworks & Libraries:** Node.js, Express.js, Spring Boot, React, MongoDB, JUnit, jQuery

**Developer Tools:** Git, Docker, Google Cloud Platform, Azure, Postman, OpenApi